

Differentiation

Directional Derivatives and the Gradient Vector

Directional Derivatives(方向导数):

Definiton: $D_{\vec{u}}f(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0+ha, y_0+hb) - f(x_0, y_0)}{h}$

ps: 这里的 $D_{\vec{u}}f(x_0, y_0)$ 被叫做沿着 $\vec{u}$ 方向上的方向向量

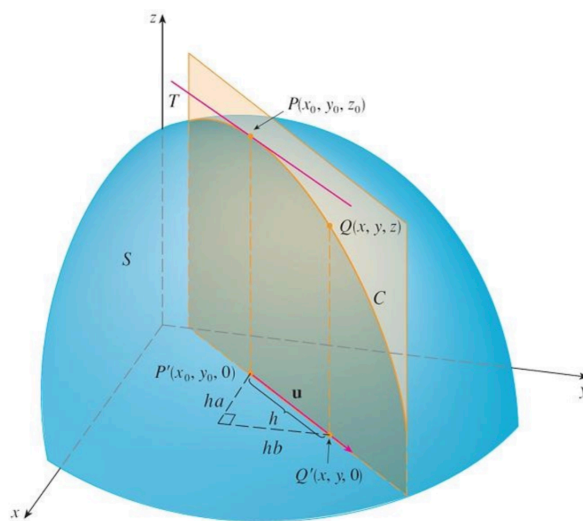
$$D_{\vec{u}}f = \nabla f \cdot \vec{u}$$

其中 $\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$ 是 gradient (梯度)。

最大值: 当方向 $\vec{u}$ 与梯度 ∇f 的方向相同时, 方向导数达到最大值。这意味着: 梯度方向是函数增加最快的方向。

最小值: 同理就是沿着相反的方向 -» [视频链接 \(方向向量与梯度的直观理解\)](#)

Suppose that we now wish to find the rate of change of z at (x_0, y_0) in the direction of an arbitrary unit vector $\mathbf{u} = \langle a, b \rangle$.



$$\mathbf{u} = \langle \cos \theta, \sin \theta \rangle$$

$$D_u f(x, y) = f_x(x, y) \cos \theta + f_y(x, y) \sin \theta$$

$$\Rightarrow \text{这个可以看出来 } D_u f(x, y) = \sqrt{f_x(x, y)^2 + f_y(x, y)^2} \cos(\theta - \varphi)$$

当 $\theta = \varphi$ 的时候这个梯度是最大的

Gradient

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = \langle f_x(x, y), f_y(x, y) \rangle$$

梯度向量 (Gradient Vector) 与切平面 (Tangent Plane) 之间的关系

1. 梯度向量 $\nabla f(x_0, y_0)$ 一定与其切平面垂直

$$F(x, y, z) = 0$$

$$\frac{\partial F}{\partial x} \frac{dx}{dt} + \frac{\partial F}{\partial y} \frac{dy}{dt} + \frac{\partial F}{\partial z} \frac{dz}{dt} = 0$$

Tangent Plane:

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

$$\text{当 } z = f(x, y)$$

$$\text{可以令: } f(x, y) - z = F(x, y, z) = 0$$

原式变成了:

$$f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0$$

Level plane(等高线)

这个也可以被认为是等值线, contour lines

$$f(x, y) = k$$

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$$

设 $\vec{v} = \langle dx, dy \rangle$ 是等高线上某一点的切向量。由于在等高线上函数值不变，该方向的方向导数必然为 0：

$$D_{\vec{v}}f = \nabla f \cdot \vec{v} = 0 \Rightarrow \nabla f \perp \text{等高线的切线}$$

Local Maximum and Local Minimum

Local Maximum : 极大值 Local Minimum: 极小值

critical points: 极致点:

Therom 1 says : A point (a, b) is called a **critical point** (or *stationary point*) of if $f_x(x, y) = 0$ and

$f_y(x, y) = 0$ or **if one of these partial derivatives does not exist.** (驻点)

Theorem 2 says : that if f has a local maximum or minimum at (a, b) , then (a, b)

is a critical point of f .

对于是否是极大值还是极小值（局部），还是马鞍点

1. 首先是 $f_x(x, y) = f_y(x, y) = 0$

2. 其次通过Hessian determinant 判断:

$$D = D(a, b) = f_{xx}(a, b)f_{yy}(a, b) - [f_{xy}(a, b)]^2$$

根据 D 的符号和 f_{xx} 的符号，可以得出以下结论：

(a) 局部极小值 (Local Minimum):

如果 $D > 0$ 且 $f_{xx}(a, b) > 0$ ，则 $f(a, b)$ 是一个局部极小值。

(b) 局部极大值 (Local Maximum):

如果 $D > 0$ 且 $f_{xx}(a, b) < 0$ ，则 $f(a, b)$ 是一个局部极大值。

(c) 鞍点 (Saddle Point):

如果 $D < 0$ ，则 $f(a, b)$ 不是极值点，而是一个鞍点（像马鞍一样，一个方向向上弯，一个方向向下弯）。

证明过程：

$$Q(h, k) = f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2$$

这里的 $o(|\mathbf{h}|^2)$ 表示比 $h^2 + k^2$ 更高阶的无穷小量。当 $|\mathbf{h}|$ 足够小时， f 的增量 Δf 的正负号完全由二次型 $Q(h, k)$ 决定。

我们可以将 $Q(h, k)$ 写成矩阵形式：

$$Q(h, k) = \begin{pmatrix} h & k \end{pmatrix} \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} \begin{pmatrix} h \\ k \end{pmatrix} = \mathbf{h}^T H \mathbf{h}$$

因此只要其行列式是大于0的并且 f_{xx} 是大于0 就说明这个点是向上凹的

Applying this theorem a second time, we have

$$\begin{aligned} D_u^2 f &= D_u(D_u f) = \frac{\partial}{\partial x}(D_u f)h + \frac{\partial}{\partial y}(D_u f)k \\ &= (f_{xx}h + f_{yx}k)h + (f_{xy}h + f_{yy}k)k \\ &= f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2 \quad (\text{by Clairaut's Theorem}) \end{aligned}$$

If we complete the square in this expression, we obtain

$$10 \quad D_u^2 f = f_{xx} \left(h + \frac{f_{xy}}{f_{xx}} k \right)^2 + \frac{k^2}{f_{xx}} (f_{xx}f_{yy} - f_{xy}^2)$$

Lagarange Multipliers

Core idea and core logic of geometric explanation: Finding the extreme value of function f when satisfying the constraint $g(x, y, z) = k$ is equivalent to finding the point where the isosurface (等值面) of f is tangent to the constraint surface. Geometric basis: At the extreme point, the gradient vector ∇f of the objective function and the gradient vector ∇g of the constraint function are parallel.

方程组：引入两个乘数 λ 和 μ ，求解以下五个方程：

- $\nabla f = \lambda \nabla g + \mu \nabla h$
- $g(x, y, z) = k$

- $h(x, y, z) = c$

解得的一般就是最后的所求极致点

Linear Approximation

$$f(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$